

The Truth Axis is a Semantically Signed Mixture, and its Arrangement Converges Across Model Families

Working note — per-category geometry, a registered cross-family falsification and its
verdict, and a contract for sparse decomposition

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Abstract

The companion paper measured that the one-dimensionality of the truth axis is knowledge-dependent; the anatomy notes measured who writes it and how. This note asks what the axis is *made of* — and whether the answer is a property of the model or of the knowledge. Grouping CounterFact by Wikidata relation and fitting per-category truth axes at the peak block, we find on three models across two families (Qwen2.5-1.5B/3B, Llama-3.2-3B): (1) the axes form a **semantically signed geometry** — language-family relations cohere positively, continent is anti-aligned with that block (confirmed independently by below-chance cross-category transfer), sport-position is near-orthogonal; (2) the FFN’s class-signed write directions decode the category far above a lexical baseline (74% vs 39% on Qwen-3B, 82% vs 42% on Llama-3B, chance 12.5%, permutation $p = 0.0099$): the network *writes* truth along category-dependent directions; (3) the geometry **reorganizes with depth** — most dramatically, the axes of *languages spoken* and *official language* start near anti-parallel at the embedding level ($\cos = -0.90$ Qwen, -0.86 Llama) and merge at the peak ($+0.76$, $+0.62$), on both families; and (4), the registered falsification (R5): the two families’ peak matrices correlate at $r = +0.77$ (Mantel label-permutation $p = 0.0009$; robust to excluding an anomalous relation), while *within* each model the early and peak arrangements are uncorrelated or anti-correlated (-0.55 , -0.18). **The arrangement law**: models share a lexical starting arrangement, abandon it, and converge on the same functional arrangement — same start, same move, same destination, different machines. The mixture’s geometry is a property of the knowledge domain, not of the recipe. We deliver the theorist’s handoff — an exported measured dictionary ($K \times d$ parameters) — and a four-criterion registered contract that any trained sparse autoencoder must satisfy, including beating the measured dictionary. Registered predictions that failed or held only partially are reported as such.

1 Scope and honesty statement

Shields as in the series: held-out cross-validation over pairs, permutation nulls, float32 decomposition gated by additive identity checks (median relative errors $1.14\text{--}1.23 \times 10^{-7}$ Qwen, 8.37×10^{-8} Llama), pre-registered predictions, no automatic verdicts. Deterministic sampling per relation was verified by exact reproduction of the $K=5$ submatrix inside the $K=8$ run. A determinism defect in the decoding fold assignment (salted string hash) was found by cross-run comparison and fixed (process-stable CRC32) before the canonical runs. The registered falsification of Section 6 was written *before* any Llama measurement of this geometry; its outcomes below include one prediction whose magnitude fell outside the registered window and one confirmed only partially — reported plainly. Open caveats: one dataset and one statement form per relation; small signed-cosine magnitudes on Qwen; a degenerate relation on Llama (Sec. 7); no bootstrap error bars yet.

2 Design

Top- K CounterFact relations by unique pairs ($K=5$ exploratory, $K=8$ for the falsification; 60 pairs each): P103 mother tongue, P1412 languages spoken, P176 manufacturer, P27 citizenship, P30 continent, P37 official language, P413 sport position, P495 country of origin. Per-category truth axes are fit at the peak block (15/16/9), oriented true = positive within their own category, so the **sign** of a cross-category cosine is information: negative means the shared direction reads the other category’s truth backwards. Measurements: signed cosine matrices at the peak and at an early block (2, the lexical surface control); the held-out transfer matrix; nearest-centroid decoding of the category from the FFN’s class-signed write direction Δf at peak+1, against the same decoder on early residual differences (the lexical baseline).

3 The signed geometry (P1–P2)

At the peak, the language-family block coheres (Qwen-3B: +0.27 to +0.76; Llama-3B: +0.55 to +0.69), continent is anti-aligned with it (Qwen -0.07 to -0.20 ; Llama -0.09 to -0.15), sport-position is near-orthogonal to everything, and country-of-origin sides with the block (strongly so on Llama, +0.51 to +0.75). The anti-alignment is confirmed by an independent statistic: transfer from the continent axis into the block falls below chance on both families (Llama: AUC 0.20–0.33); on Qwen the below-chance transfer is bidirectional, on Llama one-directional (the block’s axes read continent data above chance) — a partial replication reported as such. Within-category transfer exceeds cross on both models. The strongest single block cell: *languages spoken* and *official language* at +0.76 (Qwen) / +0.62 (Llama).

4 The reorganization (P27, and the P1412–P37 merge)

The early-block matrices are not flat: they carry shared corpus statistics (the two families’ early matrices correlate at $r = 0.93$). The structure is therefore not that depth creates arrangement from nothing, but that it *re-files* it. Citizenship sides with geography at the embedding level and with the language family at the peak, on both families. The extreme case: *languages spoken* vs *official language* start near anti-parallel ($\cos = -0.90$ Qwen, -0.86 Llama — presumably a lexical opposition in how the two templates use language tokens) and merge at the peak (+0.76, +0.62). Within each model, the early and peak off-diagonal arrangements are uncorrelated or anti-correlated ($r = -0.55$ Qwen, -0.18 Llama): the peak structure contradicts, rather than inherits, the lexical surface.

5 The FFN writes truth along category-dependent directions (P3)

Decoding the relation from the unit write direction Δf at peak+1, $K=8$: 74.2% (Qwen-3B) and 81.7% (Llama-3B) against chance 12.5% and permutation null 95th pct $\approx 15.5\%$, $p = 0.0099$ both; lexical baseline 39.2%/42.1%. Thirty-five to forty points of category structure in the write directions beyond token identity. Read with the anatomy series: the component that overwrites truth frames does so along category-dependent directions; the mixture is not only in the representation but in the writing.

6 The registered falsification and its verdict (R1–R5)

Predictions written before any Llama run of this geometry; outcomes:

- **R1** (language block on Llama, signed cosines in $[+0.15, +0.45]$): structure **confirmed**, magnitude **outside the registered window on the high side** (+0.55 to +0.69).

- **R2** (continent anti-aligned; below-chance transfer both directions): signs **confirmed**; bidirectional below-chance transfer **partial** (one direction on Llama).
- **R3** (Δf decoding ≥ 10 points over lexical, $p < 0.05$): **confirmed**, margin ~ 40 points.
- **R4** (lead: the citizenship early $\rightarrow$ peak switch replicates): **confirmed**, plus the stronger P1412–P37 merge (Sec. 4).
- **R5** (the arrangement law): **confirmed**. Off-diagonal peak matrices, Qwen-3B vs Llama-3B: Pearson $r = +0.770$ (Spearman 0.893), Mantel category-label permutation null 95th pct $+0.498$, $p = 0.0009$; robust to excluding the anomalous P176 ($r = +0.808$, $p = 0.0037$). Combined with the within-model reorganization of Sec. 4, the reading is: shared lexical start, shared functional destination, reached by the same re-filing move on different machines. The arrangement of the truth mixture is a property of the knowledge domain — a signed, truth-specific instance of cross-model representation convergence.

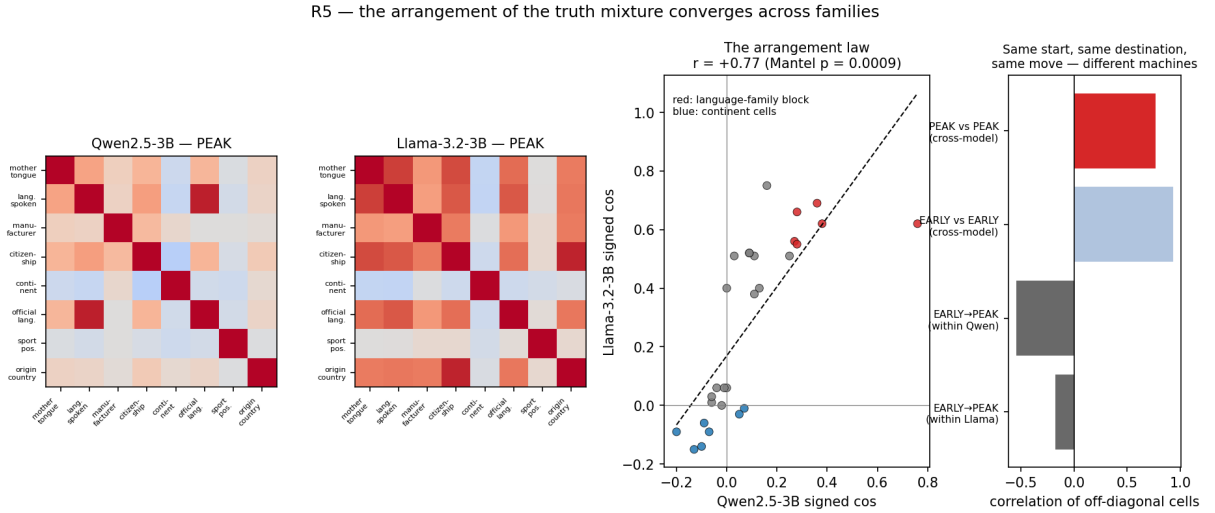


Figure 1: R5. Left: the two $K=8$ peak signed-cosine matrices. Center: their off-diagonal cells against each other ($r = +0.77$; red = language-family block, blue = continent cells). Right: the four correlations that state the law — cross-model agreement at the peak and at the embedding level, and the within-model reorganization between them.

7 Anomaly, declared

On Llama-3B, relation P176 (manufacturer) shows within-CV and incoming transfer of 1.000 from *every* axis — universal perfect separability, which suggests surface structure in that relation’s pairs rather than truth geometry. It is excluded in the R5 robustness check (which holds) and flagged for inspection before any claim rests on it.

8 The theorist’s handoff and the sparse contract

`dictionary_export.py` saves, per model, the *measured dictionary*: per-category oriented axes, the global axis, and the FFN write centroids ($[K \times d]$, $\sim$ tens of kB), with signed matrices and full metadata in one loadable bundle — a K -direction truth/category monitor whose baseline performance is quantified above. The sparse autoencoder that this series hands to engineering is justified only if, against matched nulls, it (1) aligns with the per-category axes, (2) respects

The truth axis is a semantically signed mixture: per-relation truth axes (signed cosines) and category decoding of FFN writes

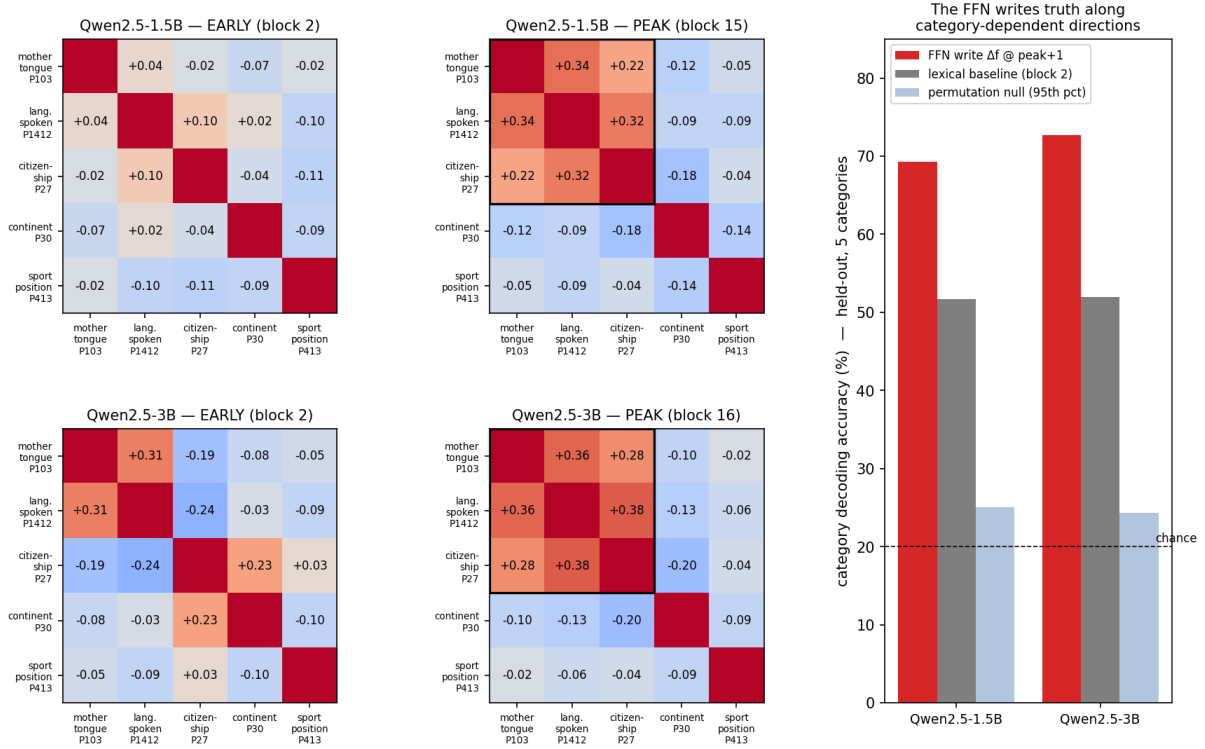


Figure 2: The exploratory $K=5$ geometry on the two Qwen models (early vs. peak signed cosines; decoding vs. lexical baseline) that motivated the registered falsification.

the semantic geography, (3) reproduces the signs — including the inverted loading on continent and the depth reorganization — and (4) **beats the measured dictionary** on class-signal reconstruction and category decoding. Either outcome is content.

9 Limitations

Two families, three models, one dataset, one statement form per relation (template syntax may contribute to the continent anti-alignment; crossed templates are the known cure). Sixty pairs per relation; two weak within-axes on Qwen attenuate their cosines; no bootstrap error bars. The R5 statistic has 28 (robustness: 21) cells; more relations would sharpen it. Llama’s uniformly higher cosines and transfer are an observation without an explanation here. The arrangement law is two families deep — a third family would promote it from law-candidate to law.

10 Reproducibility

```
python categories.py --model Qwen/Qwen2.5-3B --peak 16 --write-layer 17 --k-relations 8
python categories.py --model meta-llama/Llama-3.2-3B --peak 9 --write-layer 10 --k-relations 8
python arrangement_law.py # R5 + controls from the canonical matrices
python dictionary_export.py --model Qwen/Qwen2.5-3B --peak 16 --write-layer 17
```

Scripts `categories.py`, `dictionary_export.py`, `arrangement_law.py` in `src/`; figures `arrangement_law.png`, `categories_geometry.png`. License: CC BY 4.0.

References

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